



Biochip Course

Impedance Measurement

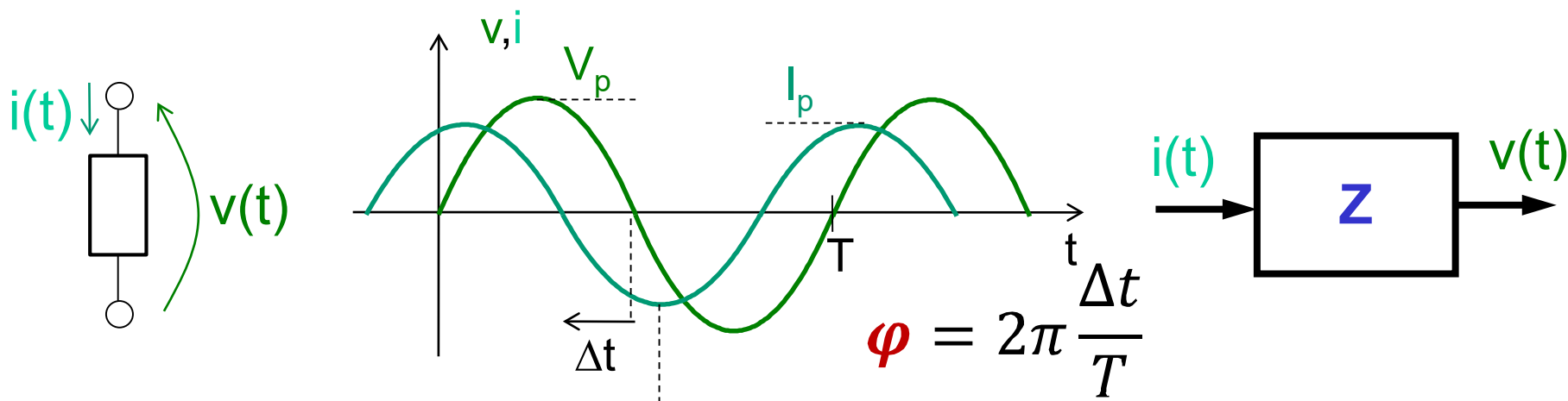
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a.y. 2022/2023

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Definition



$$\mathbf{Z}(\omega) = \frac{|V_p|}{|I_p|} e^{j\varphi}$$

Impedance [Ohm]

- Complex quantity
- Change with frequency

$$\mathbf{Y}(\omega) = \frac{|I_p|}{|V_p|} e^{-j\varphi} = \frac{1}{Z(\omega)}$$

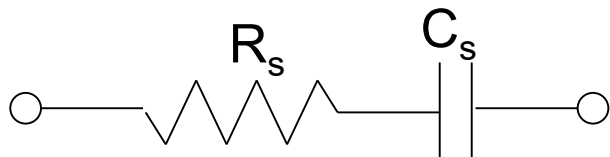
Admittance [Siemens]



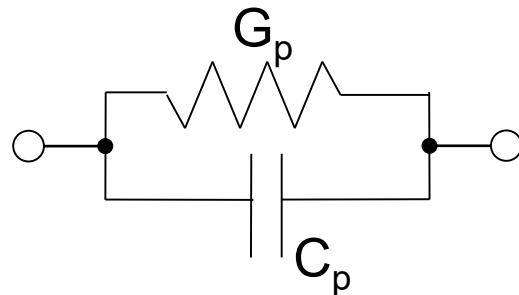
Definition

Express impedance in terms of **resistance** and **capacitance**:

$$Z(\omega) = R_s + \frac{1}{j\omega C_s}$$



$$Y(\omega) = G_p + j\omega C_p$$



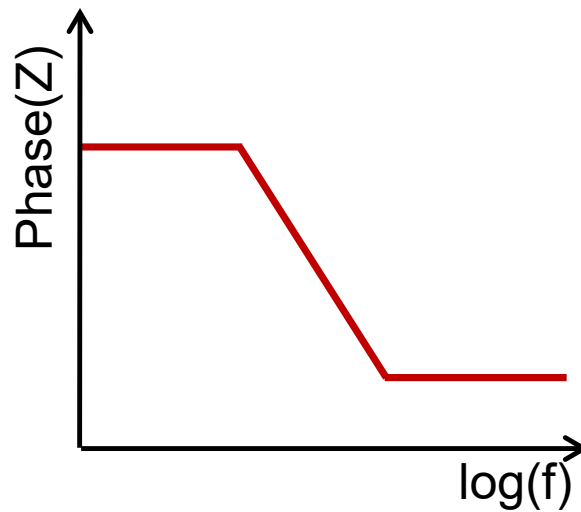
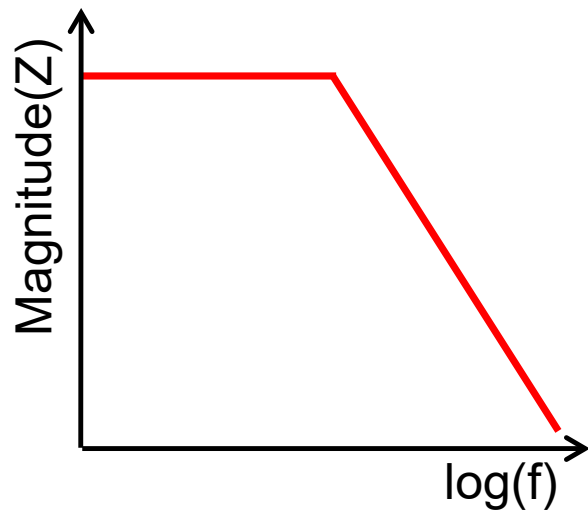
- In general $R_s \neq 1/G_p$ and $C_s \neq C_p$, all frequency-dependent
- $Z(\omega)$ or $Y(\omega)$ give complete description of a linear network
- $\text{Re}\{Z\}, \text{Re}\{Y\} \rightarrow$ **energy dissipation** ($4kT\text{Re}\{Z\}$, $4kT\text{Re}\{Y\}$)
- $\text{Im}\{Z\}, \text{Im}\{Y\} \rightarrow$ **energy storage**



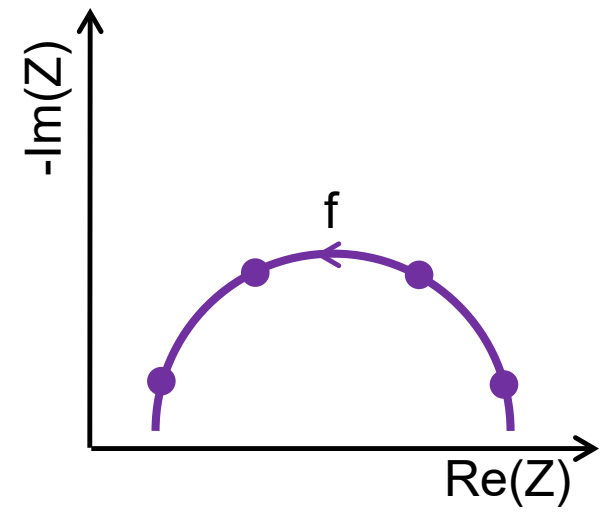
Plotting Impedance Spectra

Two alternative but **equivalent** ways to present impedance spectra:

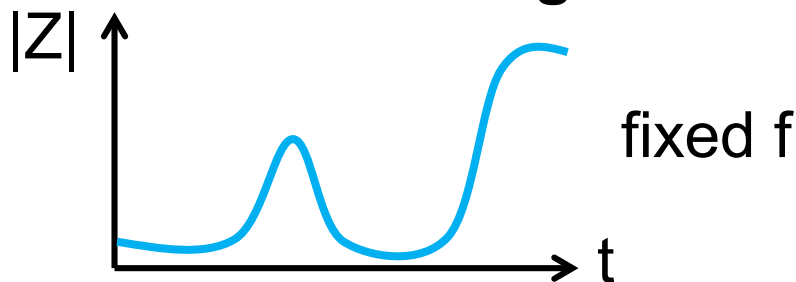
Bode Plots



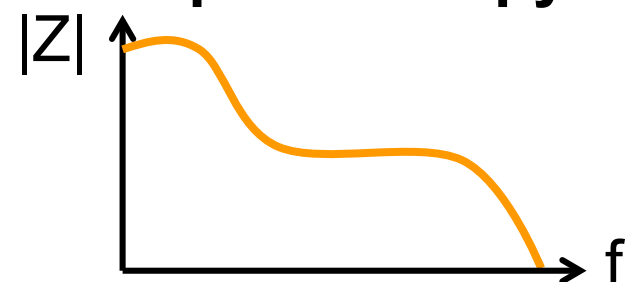
Cole-Cole Plot



Time tracking



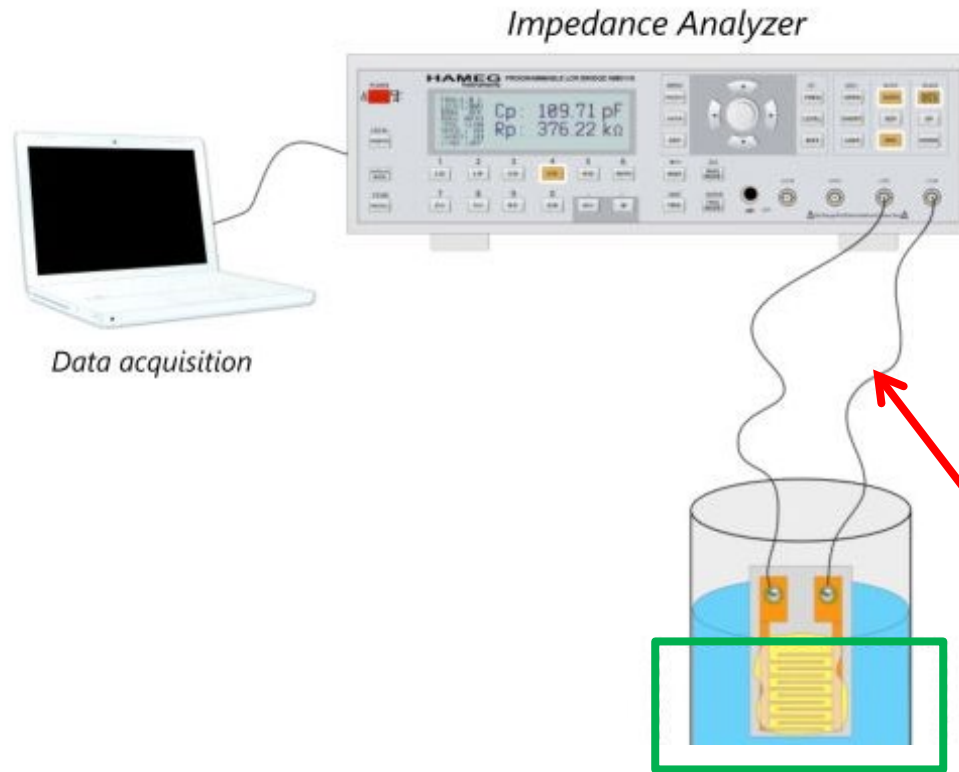
Spectroscopy





Microscale Impedance Spectroscopy

Microfabriaction allows the realization of **microelectrodes** to interface with microscale samples (such as biological cells).



The impedance of the micro sample **scales** with the size

while

the parasitic impedance of the macroscopic **connections** stays fixed.

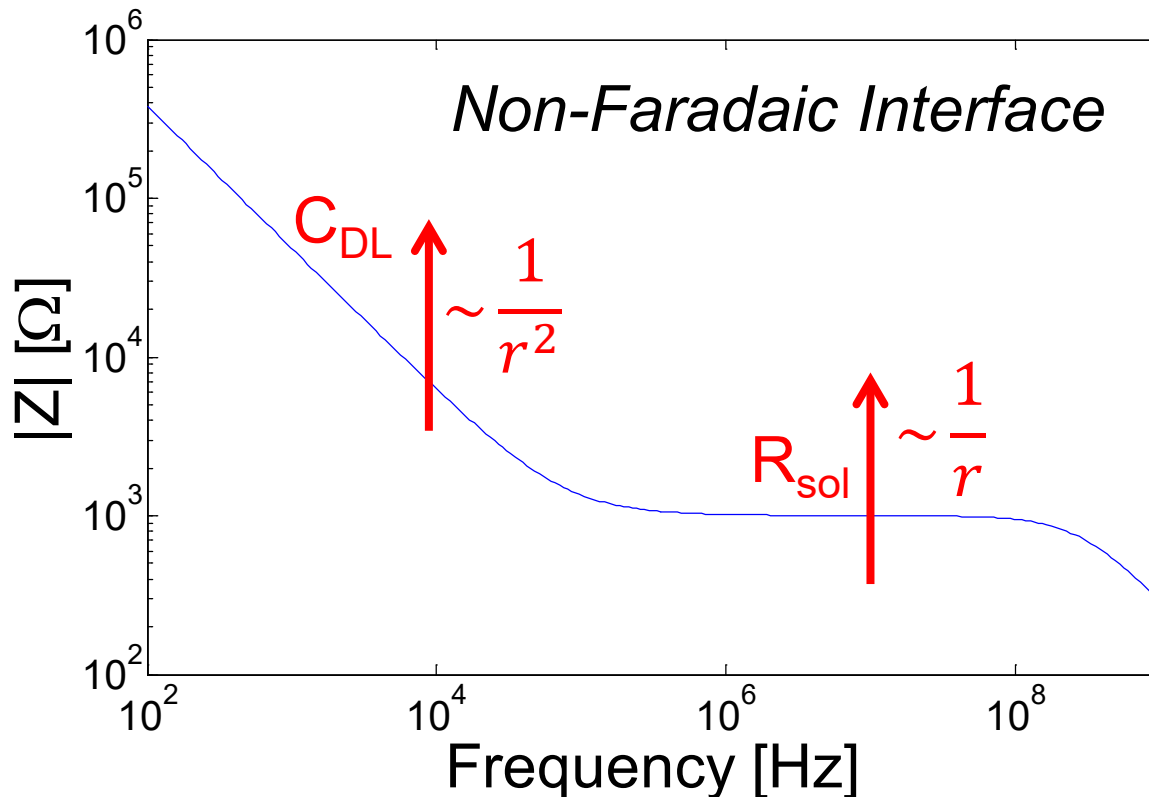
Impedance of Microelectrodes

Planar disk microelectrode in ionic solution (PBS):

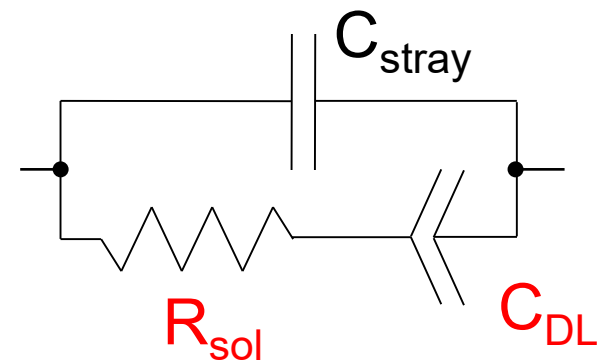


$$R_{sol} = \frac{\rho}{4r}$$

$$C_{DL} = 0.1 \frac{pF}{\mu m^2} \cdot 2\pi \cdot r^2$$



$$Z = R_{sol} + \frac{1}{sC_{DL}}$$

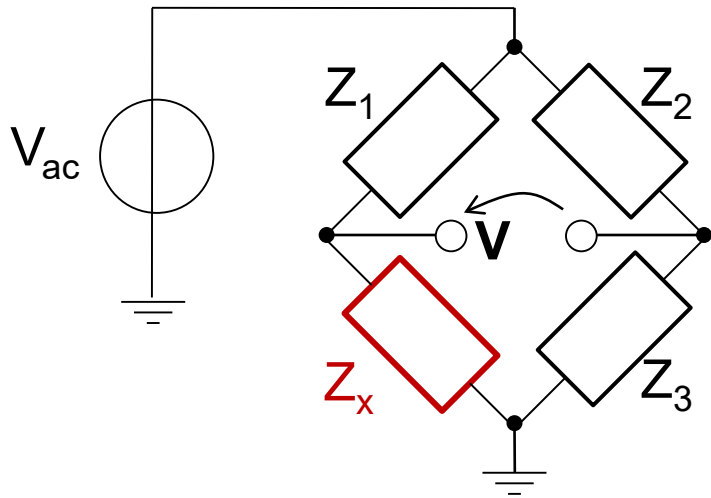


Analyze and compare from the point of view of noise for biochips

- **Bridge**
- **Ratiometric**
- **Current sensing**
- FFT-based method
- Resonating techniques
- Charge-based measurements



Balancing Bridge: Working Principle



- Z_1, Z_2, Z_3 known and variable (switches)
- V_{ac} sinusoidal

$$V = V_{ac} \left(\frac{Z_x}{Z_1 + Z_x} - \frac{Z_3}{Z_2 + Z_3} \right)$$

Balanced for $V = 0 \Rightarrow Z_x = Z_3 \frac{Z_1}{Z_2}$



Bridge Pros and Cons

Pros:

- Good accuracy (no active stages, depends on the accuracy of the reference impedances)
- Voltage reader operates always with $V \approx 0V$
- Common mode rejection

Cons:

- Requires several switches
- Slow balancing routine
- Not very convenient for spectroscopy

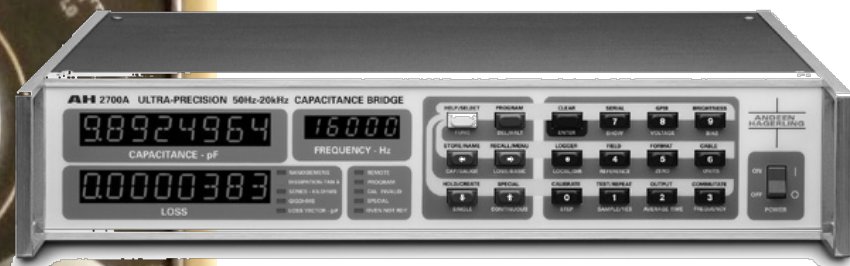


Examples of Commercial Instruments

GR 1650-A (1957) ...fully manual



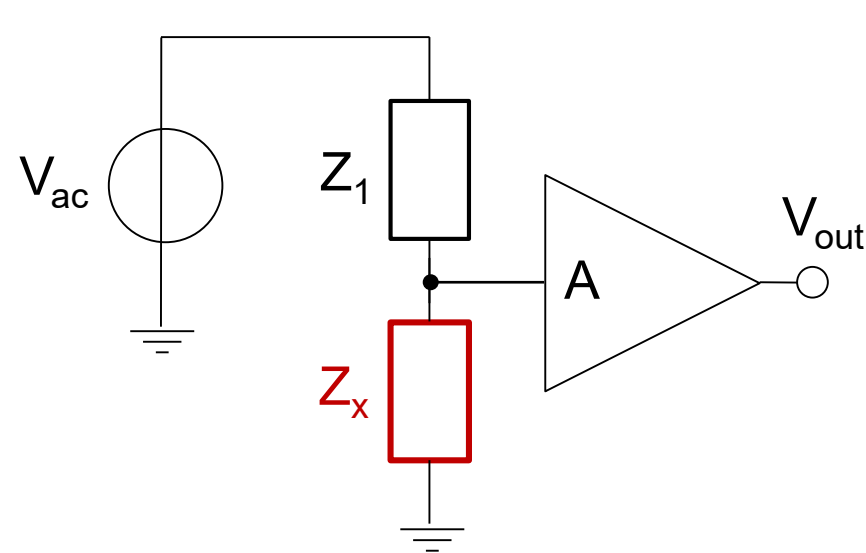
Andeen-Hagerling
AH 2700A
50Hz-20kHz
Capacitance Bridge



0.5aF resolution



Ratiometric: Half Bridge



$$V_{out} = A V_{ac} \frac{Z_x}{Z_1 + Z_x}$$



$$Z_x = Z_1 \frac{V_{out}}{A V_{ac} - V_{out}}$$

- V_{out} depends on the impedance ratio
- Z_1 has to be accurate:
 - $Z_x \gg Z_1$: $A V_{ac} - V_{out} \approx 0$
 - $Z_x \ll Z_1$: $V_{out} \approx 0$

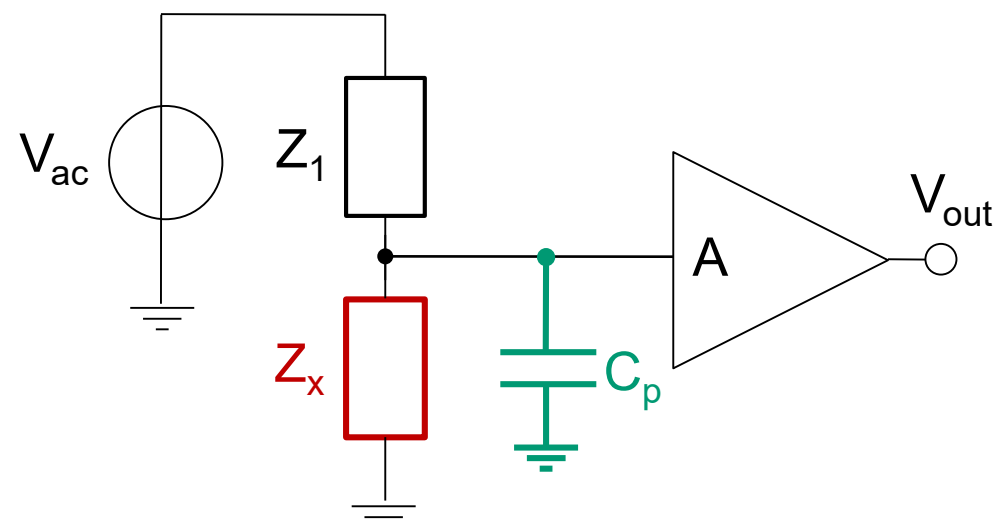


$$Z_1 \sim Z_x$$

Difficult matching



Ratiometric: Effect of Stray Cap.

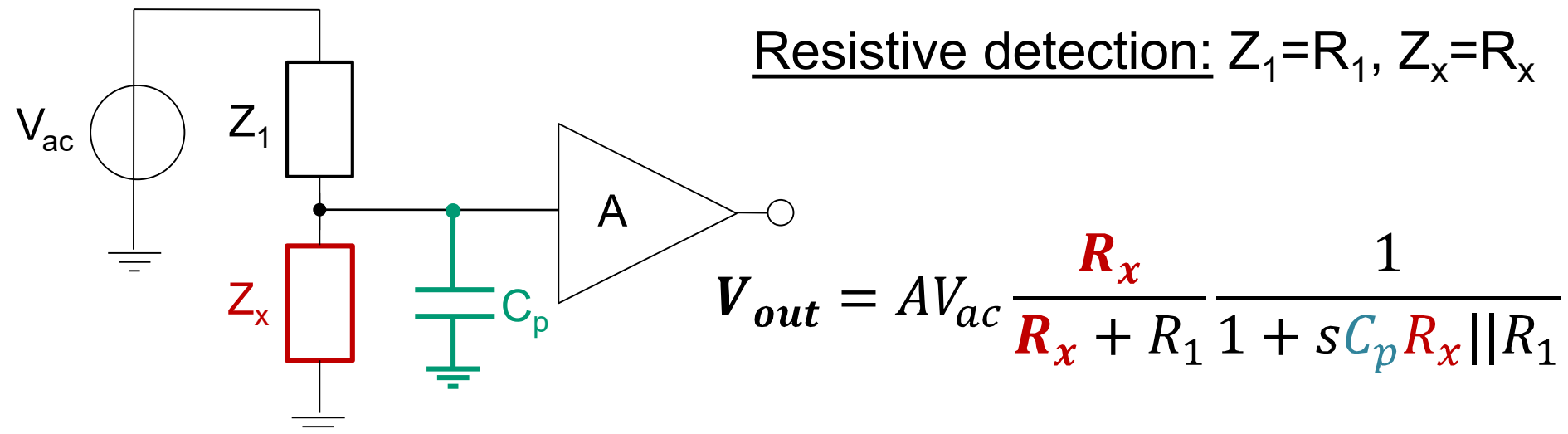


Capacitive detection: $Z_1 = 1/sC_1$, $Z_x = 1/sC_x$

$$C_x = C_1 \frac{AV_{ac} - V_{out}}{V_{out}} - C_p$$

C_p reduces the accuracy!

Ratiometric: Effect of Stray Cap.



• $\omega \ll 1/C_p R_x || R_1$:

$$R_x \cong R_1 \frac{V_{out}}{AV_{ac} - V_{out}}$$

• $\omega \gg 1/C_p R_x || R_1 \rightarrow R_x$ shunted by C_p !

Example: a cube of intrinsic Si ($\sim 1\text{k}\Omega\text{ cm}$), side = 50nm

$\rightarrow R_x = 200\text{M}\Omega$, cut-off frequency = 160Hz ($C_p = 5\text{pF}$)

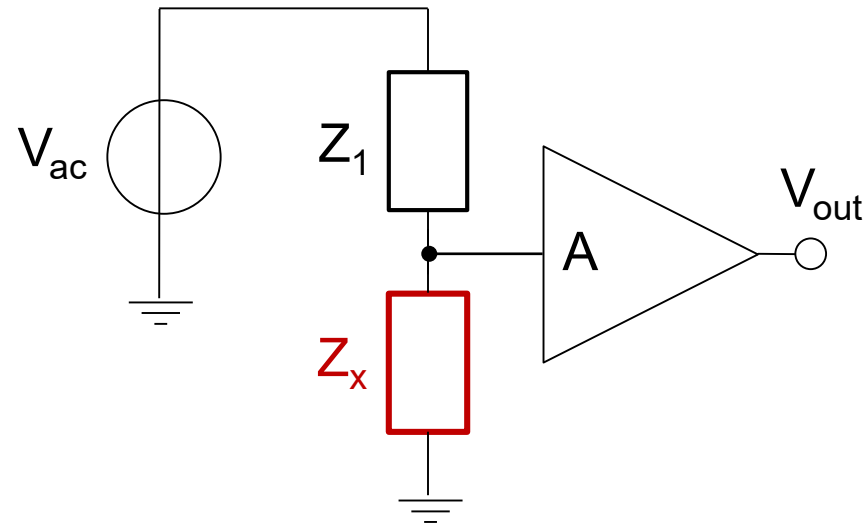


Pros:

- Independent of the impedance
- One terminal can be grounded

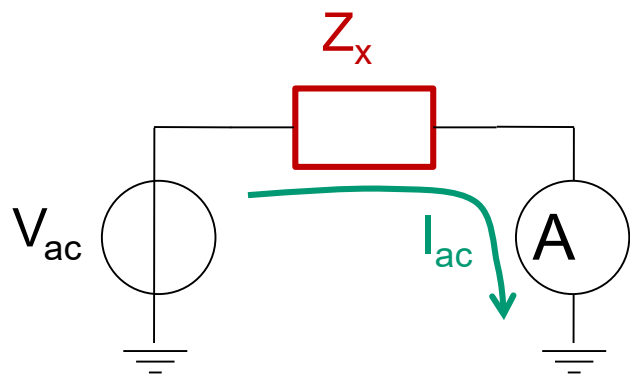
Cons:

- Z_1 must match Z_x
- Need for phase-sensitive detection for complex impedance
- Critical impact of stray capacitance:
 - Limits the bandwidth in resistive sensing
 - Reduces the accuracy in capacitive sensing





Current Sensing

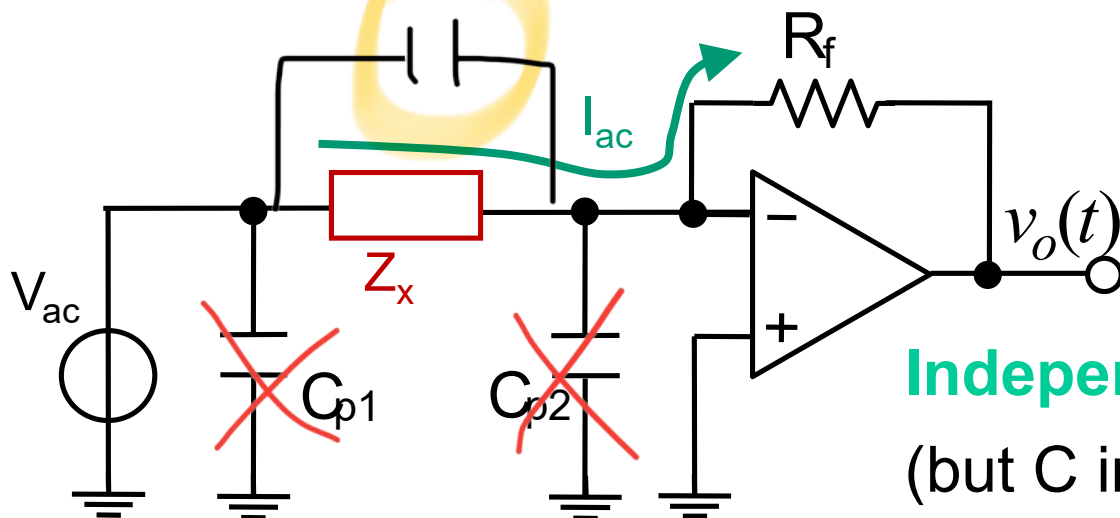


$$Z(\omega) = \frac{V_{ac}(\omega)}{I_{ac}(\omega)}$$

Phase-sensitive detector to improve the SNR and obtain the phase

I not independent

Current readout: transimpedance amplifier

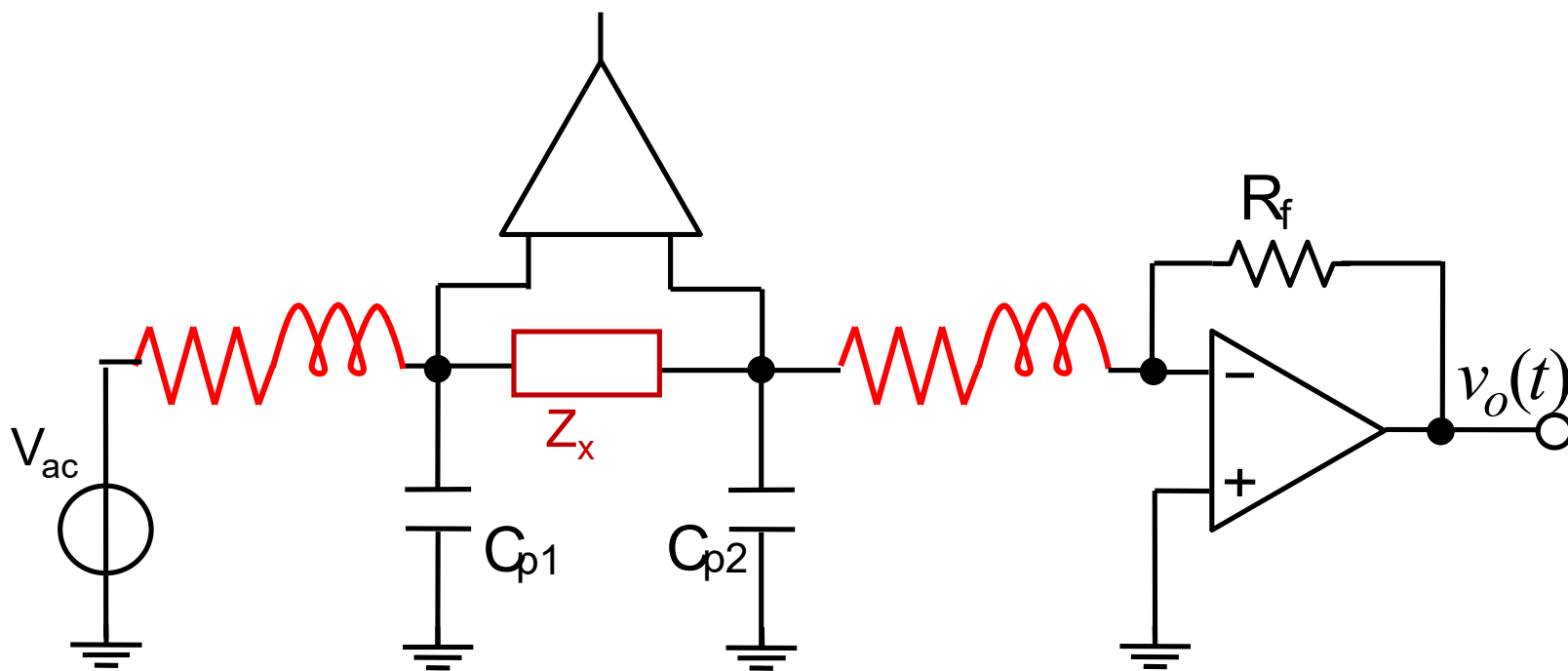


Independent of C_p cables!
(but C in parallel to Z_x)



Current Sensing

In macro-scale setups often it is adopted the **4-wire configuration**:

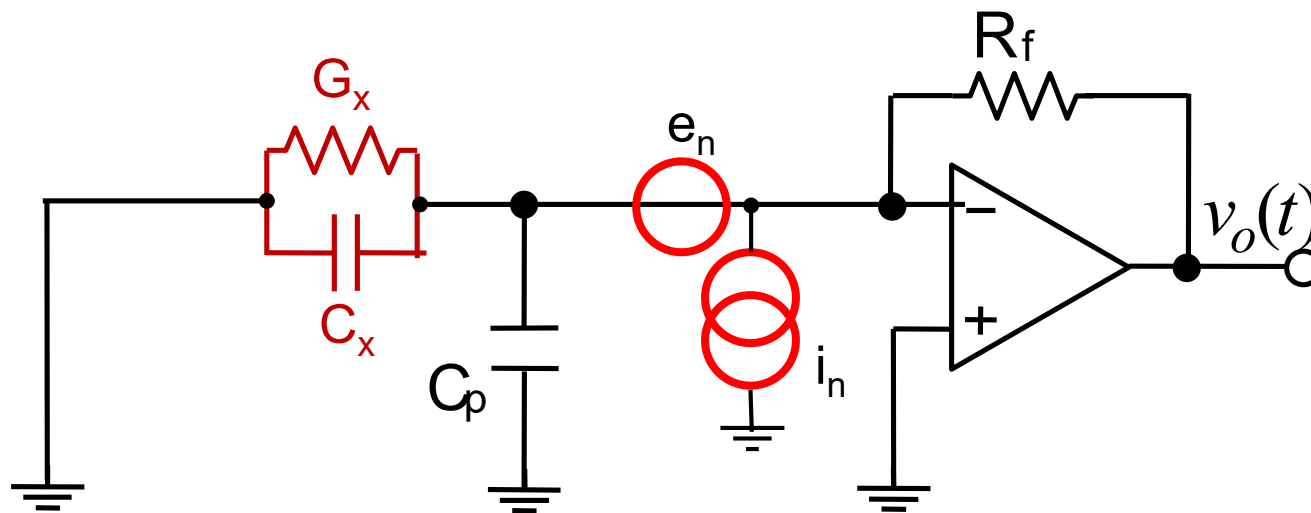


$$L = 100\text{nH}, f = 10\text{MHz} \rightarrow R = 6.2\Omega$$

Nanoscale $I_{\text{max}} \approx \text{tens of } \mu\text{A}$ (Joule dissipation)

$\rightarrow$ typically **inductances are negligible**

Current Sensing: Noise Analysis



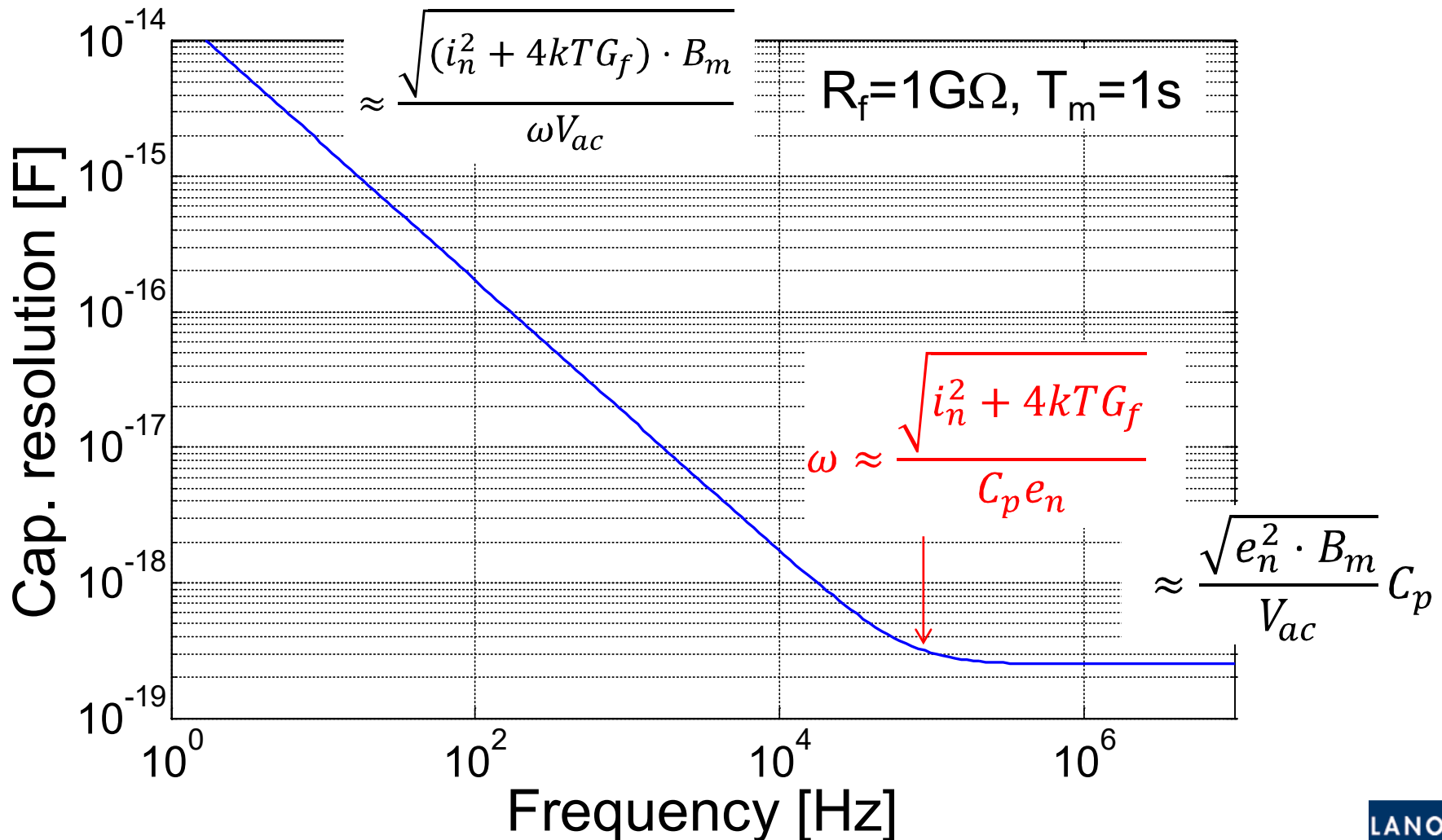
$$S_i = \bar{i}_n^2 + 4kT(G_f + G_x) + \bar{e}_n^2 \omega^2 (C_x + C_p)^2 + \bar{e}_n^2 (G_x + G_f)^2$$

$$V_{ac} \Delta G_{xmin} = \sqrt{S_i B_m} \quad \Rightarrow \quad \Delta G_{xmin} = \frac{\sqrt{S_i B_m}}{V_{ac}}$$

$$V_{ac} \omega \Delta C_{xmin} = \sqrt{S_i B_m} \quad \Rightarrow \quad \Delta C_{xmin} = \frac{\sqrt{S_i B_m}}{\omega V_{ac}}$$

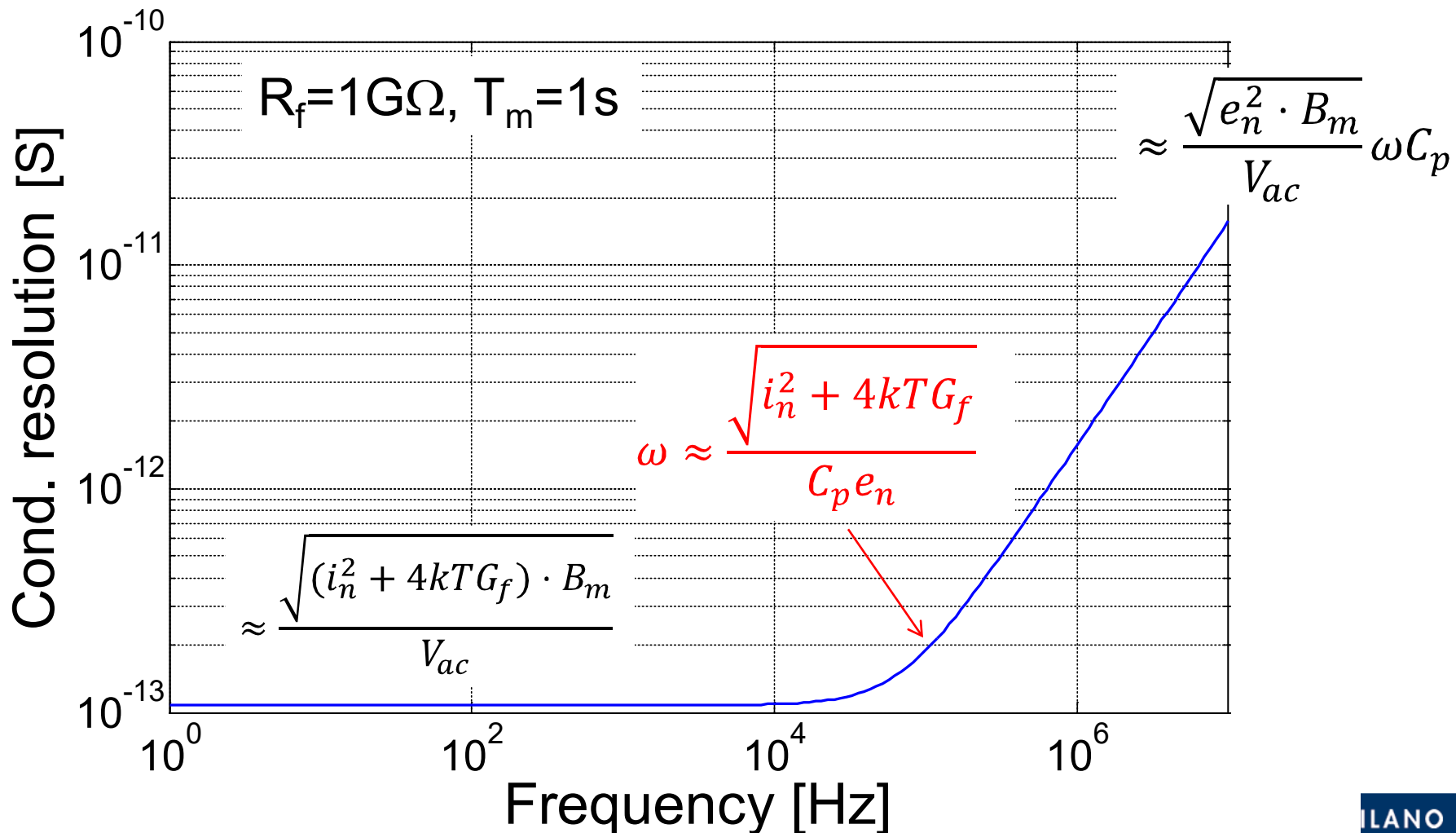
Current Sensing: Noise Analysis

$$e_n = \frac{5nV}{\sqrt{Hz}}, i_n = \frac{10fA}{\sqrt{Hz}}, V_{ac} = 100mV, C_p = 5pF, C_x \ll C_p$$



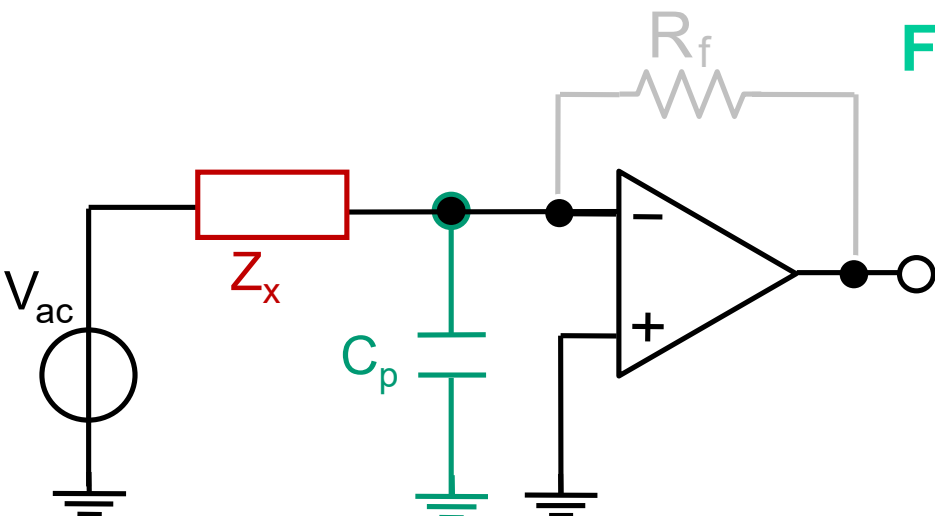
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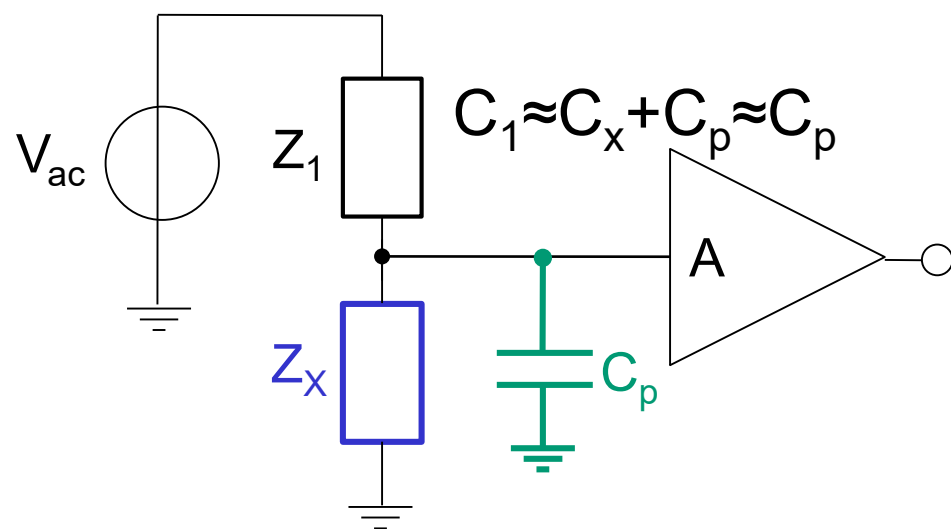


Resolution Comparison



Feedback does NOT alter SNR:

$$\Delta C_x \approx \frac{\sqrt{e_n^2 \cdot B_m}}{V_{ac}} C_p$$



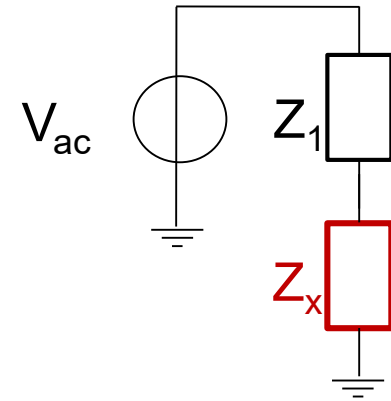
The smaller the C_p the better the resolution



Comparison

Ratiometric:

- C_p limits bandwidth and accuracy
- No control of the voltage applied to Z_x



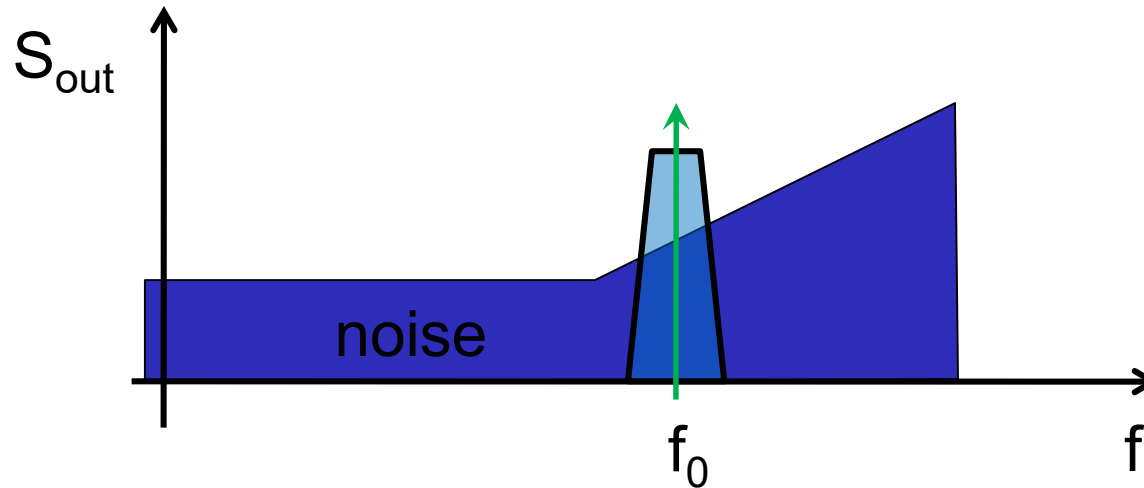
Current sensing:

- Independent of C_p (wide-band opamp)
- Precise control of the voltage applied
- Need to access both terminals of Z_x
- Loop stability depends on Z_x (but at the nanoscale dominated by stray capacitance $\approx$ known)

In terms of resolution they are **equivalent**



In both cases a sinusoidal excitation at f_0 is applied:

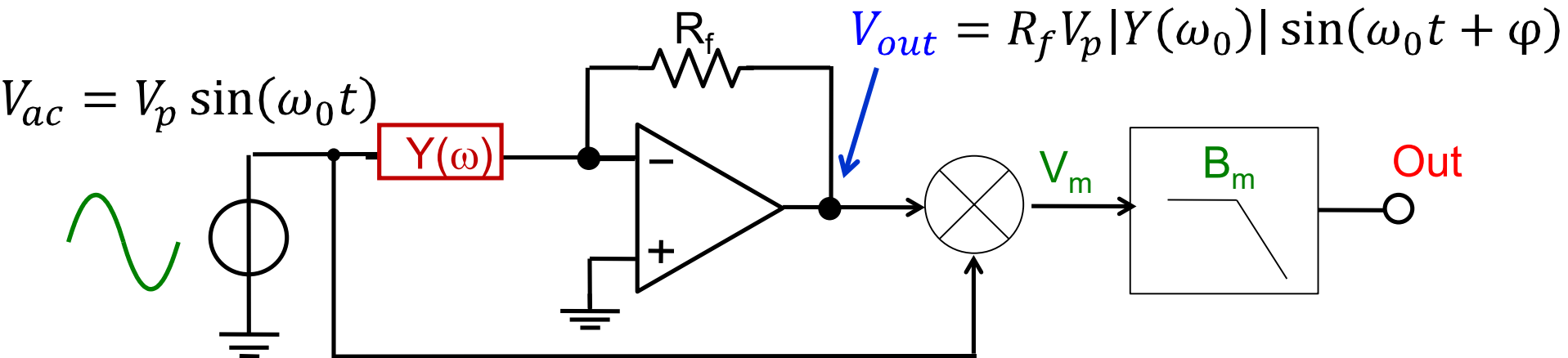


A **band-pass filter** is required to maximize the SNR:

- Very selective ($T_m=1s$, $f_0=1MHz \rightarrow \Delta f/f_0=10^{-6}$)
- Variable frequency (large spectrum sub-Hz – MHz)

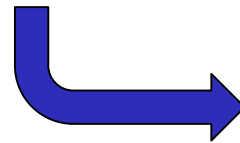


Lock-In: Working Principle



$$V_m = V_{ac}(t) \times V_{out}(t) = R_f V_p^2 |Y(\omega_0)| \sin(\omega_0 t) \times \sin(\omega_0 t + \varphi)$$

$$V_m = R_f V_p^2 |Y(\omega_0)| \frac{\cos(\varphi) - \cos(2\omega_0 t + \varphi)}{2}$$



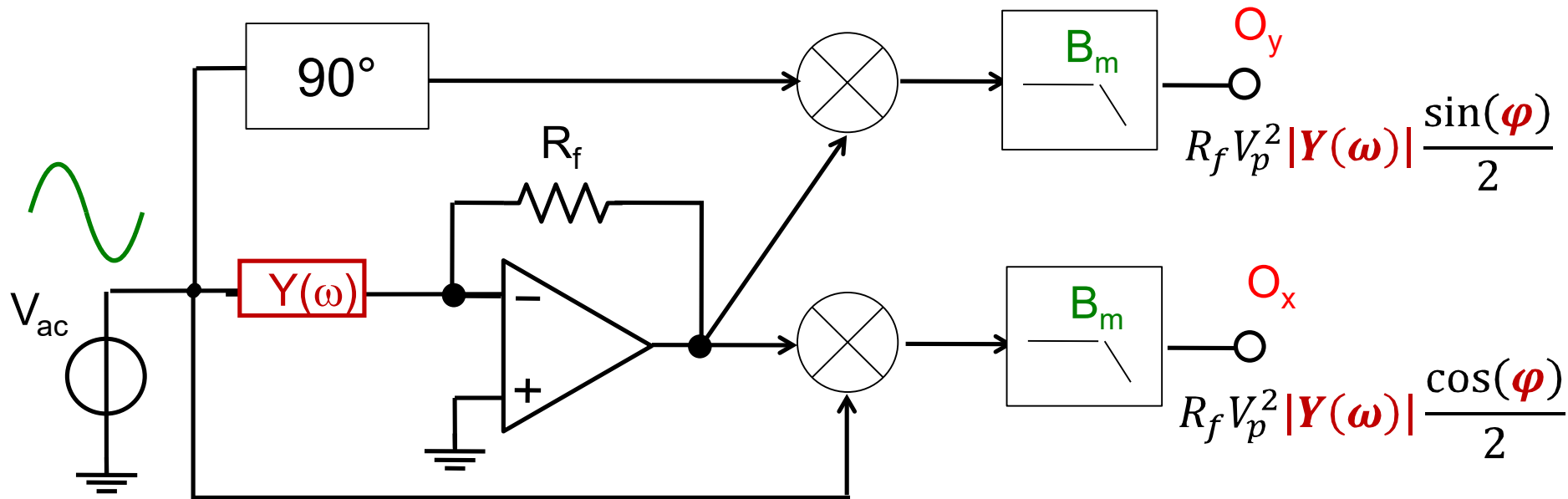
$$Out = R_f V_p^2 |Y(\omega_0)| \frac{\cos(\varphi)}{2}$$

Out DC signal corresponds to the admittance (in-phase)

The **averaging time** is set by the low-pass filter

Lock-In: Complete Impedance

Two multipliers to obtain both Re and Im:

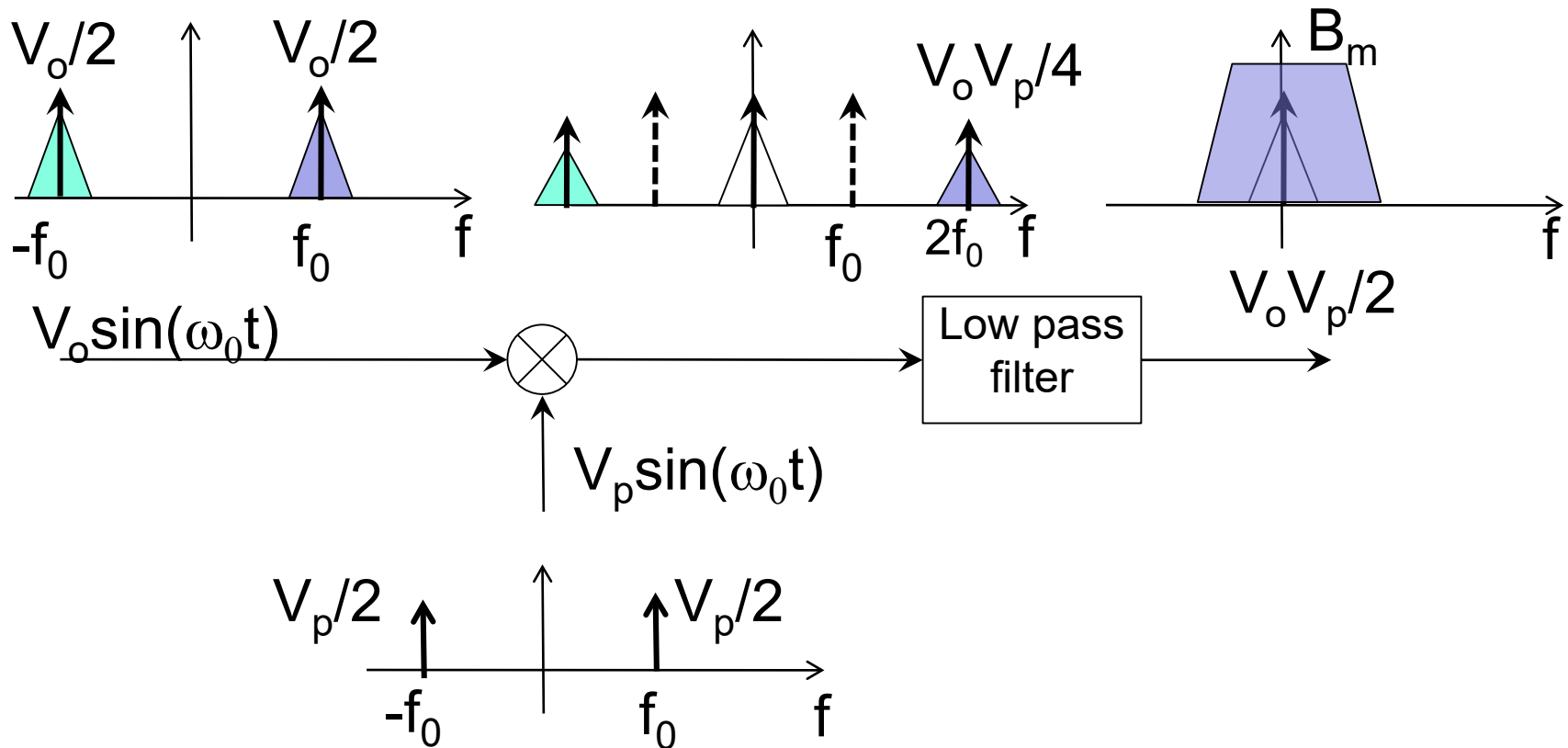


$$|Y(\omega_0)| = \frac{2 \sqrt{O_x^2 + O_y^2}}{R_f V_p^2}$$

$$\varphi = \text{atan} \frac{O_y}{O_x}$$



Lock-In: Frequency Domain

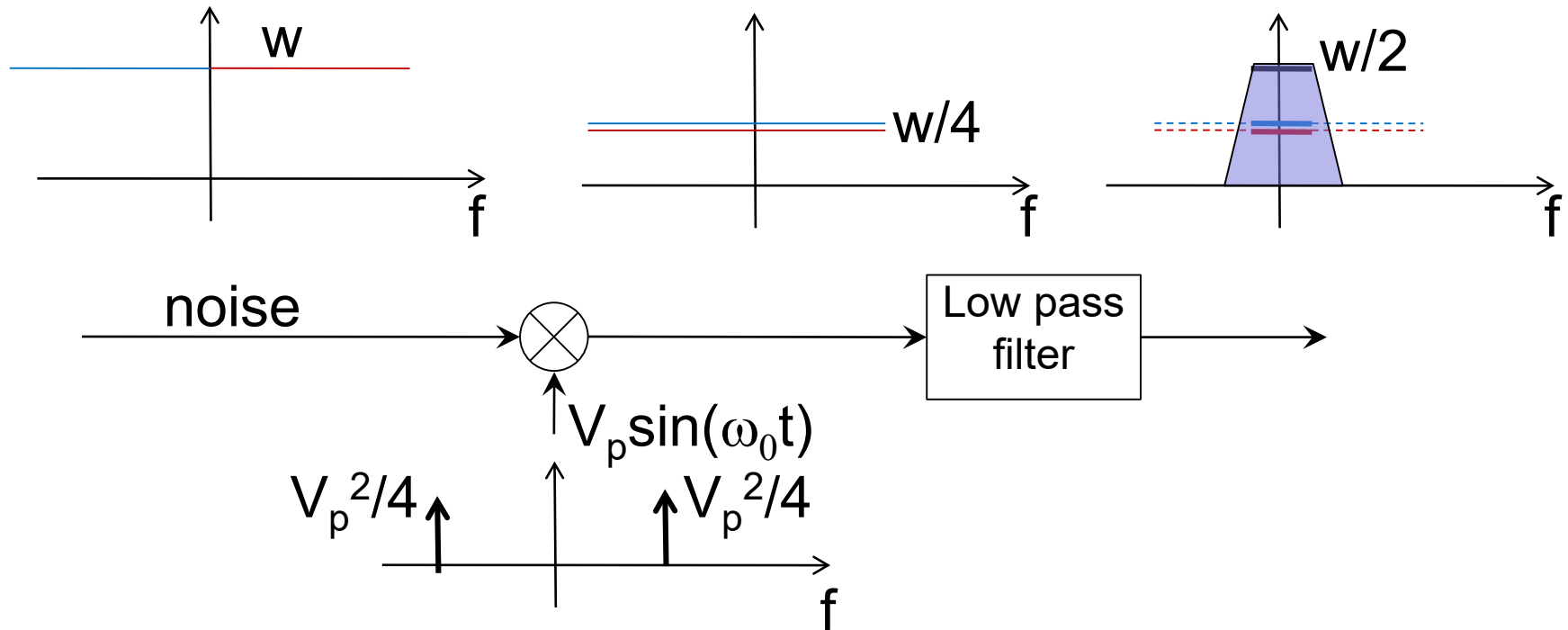


Half of the signal in base band, half is filtered at $2f_0$



Lock-In Noise Analysis White Noise

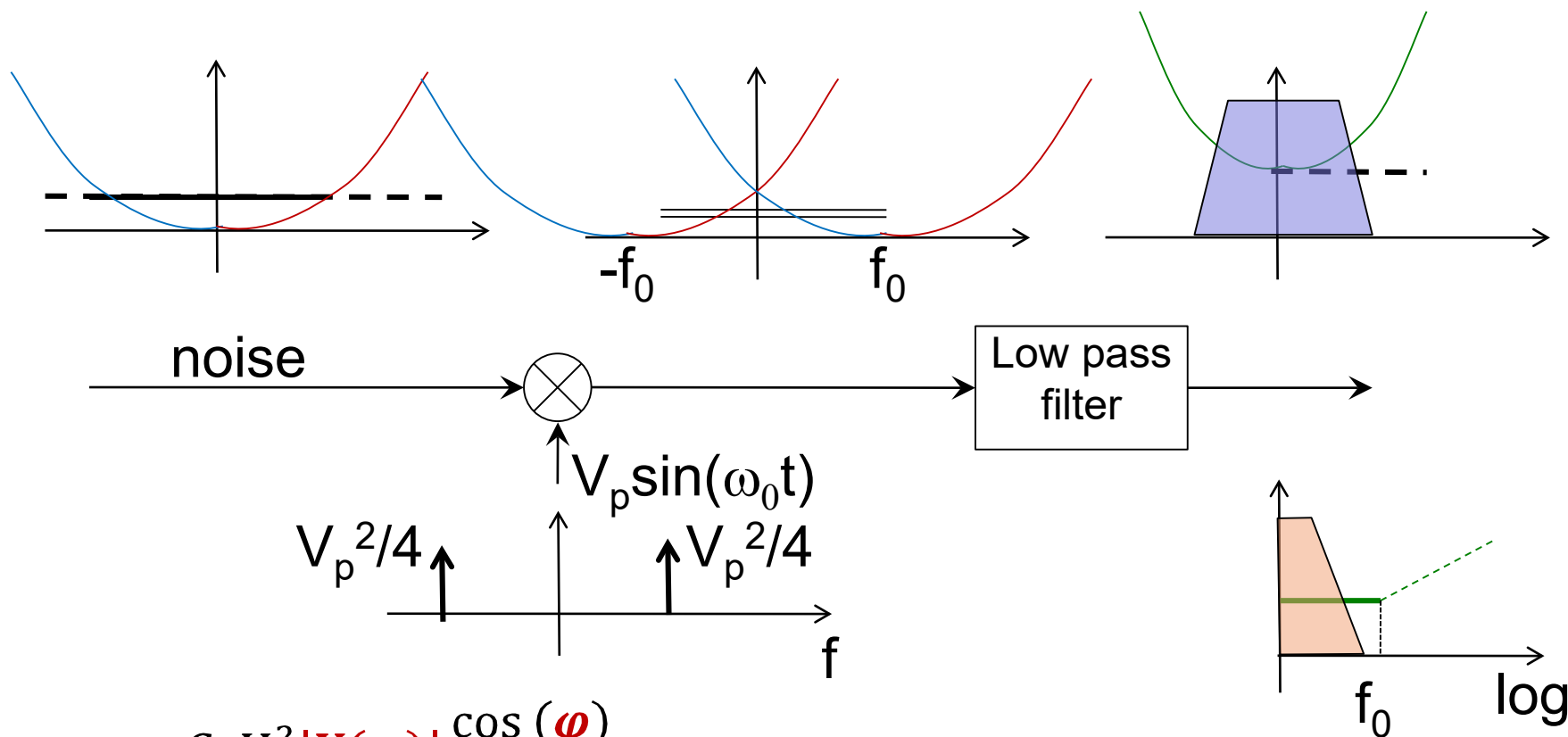
Uncorrelated noise: **summing the power**



$$\frac{S}{N} = \frac{G_R V_p^2 |Y(\omega)| \frac{\cos(\varphi)}{2}}{\sqrt{V_p^2 G_R^2 \frac{l_n^2}{2} B_n}} = \frac{V_p |Y(\omega)| \cos(\varphi)}{\sqrt{2 l_n^2 B_n}}$$



Lock-In: Non-White Noise



$$\frac{S}{N} \cong \frac{G_R V_p^2 |Y(\omega)| \frac{\cos(\varphi)}{2}}{\sqrt{V_p^2 G_R^2 \frac{\bar{l}_n^2(f_0)}{2} B_n}} = \frac{V_p |Y(\omega)| \cos(\varphi)}{\sqrt{2 \bar{l}_n^2(f_0) B_n}}$$



The lock-in filter ***is not*** a band-pass centered at f_0 :

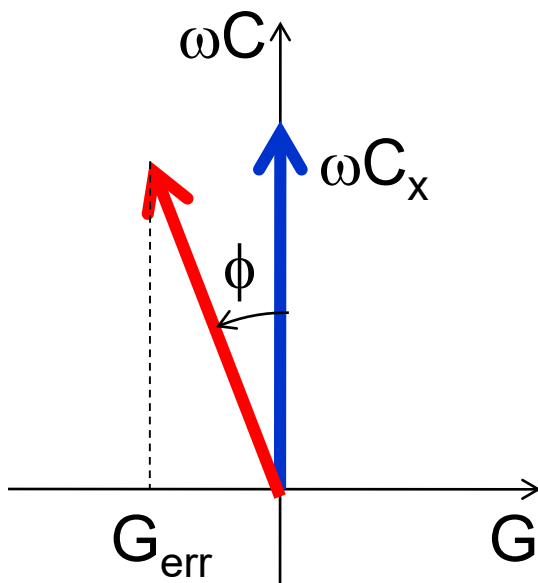
- The modulation *whiten* the noise spectrum in base band at the value sampled at f_0
- The order of the filter is not critical for the noise (a first order is ok even with non-white noise)

but

- The filter has to properly cut the $2f_0$ component



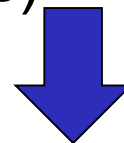
Amplifiers and connections introduce errors in amplitude and phase. Phase is more critical



Example:

Capacitor 1pF at 1MHz

Phase error 0.6° (a pole distant two decades)



$$G_{err} = \omega C_x \sin(\phi) = 6 \cdot 10^{-8} \text{ S} \\ (16 \text{ M}\Omega)$$

Ideal calibration: with a known sample (amplitude and phase)

Simplified calibration: open circuit (phase only)

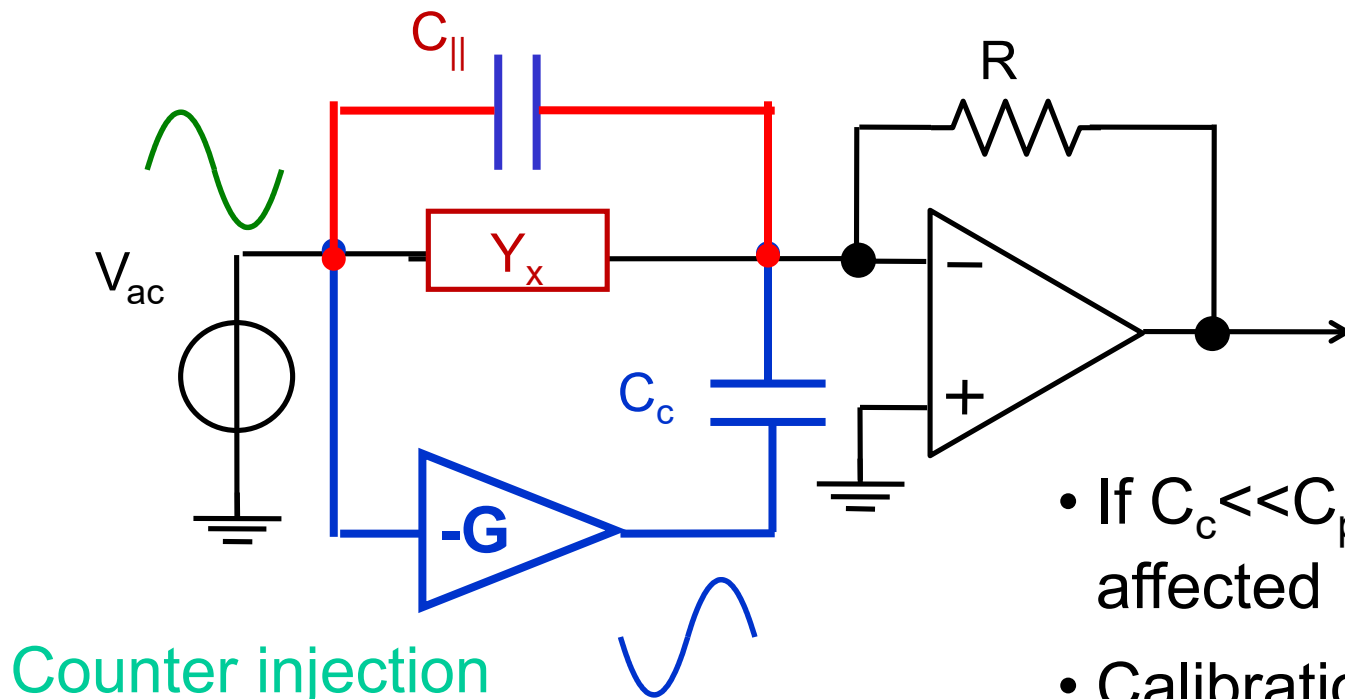


Compensation

The stray parallel capacitance $C_{||}$ may:

- saturate the front-end or gain stages
- require ADC with large bit number

An **active capacitance compensation** can be useful:



- If $C_c \ll C_p$ the noise is not affected
- Calibration required

Impedance spectroscopy with lock-in requires a separate measurement for each frequency → **long time**

Alternative **time-domain approach**:

- Apply a non-monochromatic stimulation V_{ac} (white noise...)
- A/D conversion of the transimpedance output
- Calculate the DFT of the current signal

$$Y(\omega_i) = \frac{DFT[I_{out}]_i}{DFT[V_{ac}]_i} \quad \begin{array}{l} N_c \text{ points in frequency} \\ \Delta f = f_s/N_c \end{array}$$

$$f_{\max} = 1\text{MHz}, f_{\min} = 1\text{Hz} \rightarrow f_s \approx 5\text{MHz}, N_c \approx 10\text{MSamples}$$

Noise analysis:

$$Y_{mis}(\omega_i) = \frac{DFT[I_{out}^{id}]_i + DFT[noise]_i}{DFT[V_{ac}]_i} = Y_{id}(\omega_i) + \overbrace{\frac{DFT[noise]_i}{DFT[V_{ac}]_i}}^{\text{As before}}$$

Maximize $DFT[V_{ac}]_i$ while limiting V_{ac} to avoid non-linearity

Optimal solution: do not waste energy

add only the sinusoids (not harmonic) at the desired frequencies

$$V_{ac}(t) = \sum_{i=1}^N A_i \sin(\omega_i t + \varphi_i)$$



Pros:

- Many measurements in parallel reducing the total spectroscopy time

Cons:

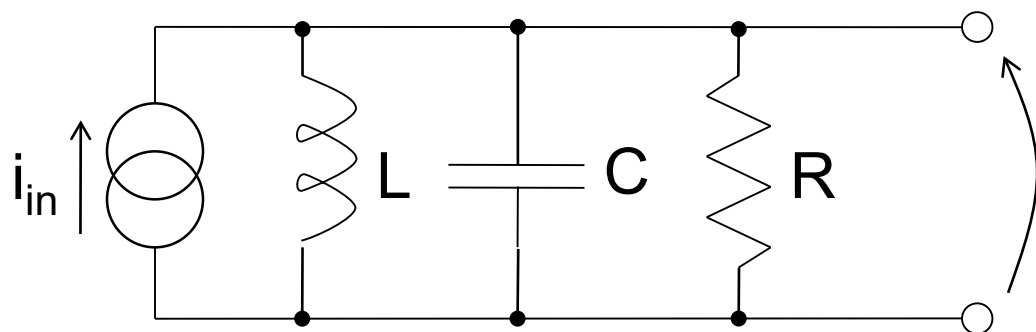
- More complicated:
 - FFT over a large spectrum is computationally heavy
 - Need for an arbitrary waveform generator
- Considering the same V_{ac}^{rms} , it is more noisy
- System linearity is critical



Resonating Techniques

Resolution limits set by C_p and e_n .

It is possible to improve the sensitivity by exploiting the resonance of a passive network?



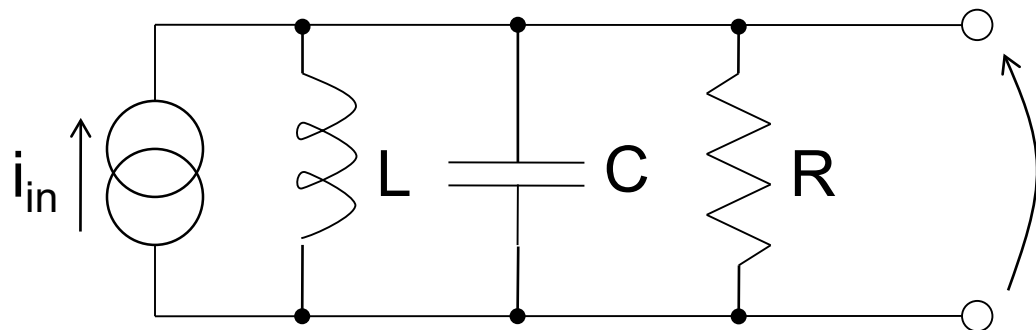
$$V_o(s) = i_{in} \frac{sL}{1 + s\frac{L}{R} + s^2LC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

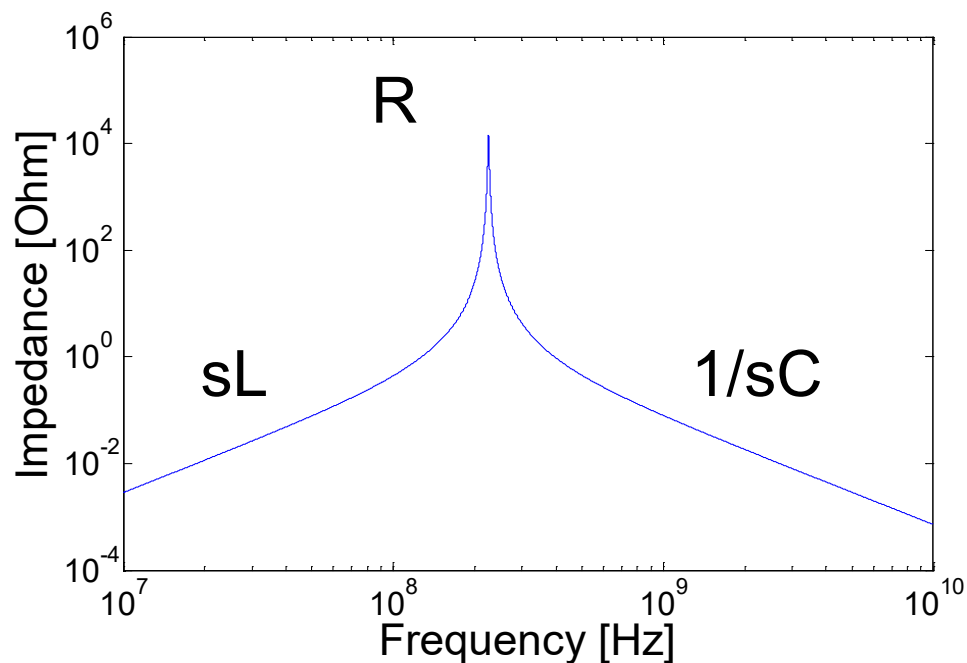
$$Q = \omega_0 CR = \frac{R}{\omega_0 L} = 2\pi \frac{E_{stored}}{E_{dis} \times ciclo}$$



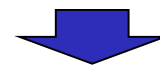
Resonating Techniques



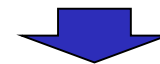
$$V_o(s) = i_{in} \frac{sL}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}}$$



$$C \approx \text{pF}, L \approx 100\text{nH}$$



$$f_0 \approx \text{hundreds of MHz}$$
$$Q \approx 100$$



$$R \approx 100\text{k}\Omega$$

In nanoscale domain used to probe small capacitance variations

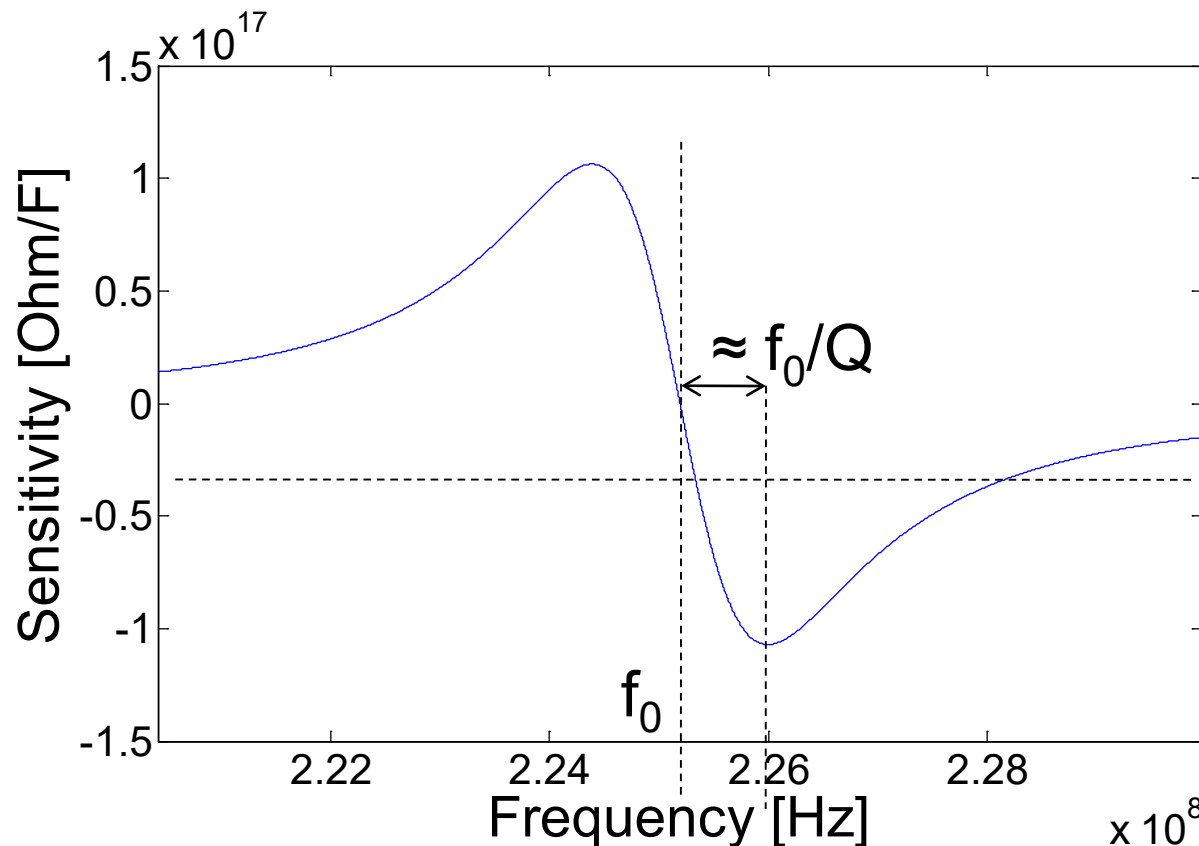


Resonating Techniques

Method: (Palmer, Denlinger, Kawamoto, *RCA Rev.* 43, p. 194 (1982))

- Apply i_{in} at a fixed frequency
- Measure the output voltage

Operating frequency:



Optimal $\approx f_0 + f_0/Q$

At $f_0 + f_0/Q$:

$$\text{Sensitivity} = \frac{\partial \left| \frac{V_{out}}{I_{in}} \right|}{\partial C} \cong \frac{4\sqrt{5}}{25} \frac{Q}{C} R \quad [\Omega/F]$$

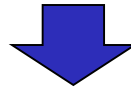
$$\text{Noise} = \left(\frac{4kT}{R} + \overline{i_n^2} \right) |Z|^2 + \overline{e_n^2} \cong 4kT \frac{4}{5} R + \overline{e_n^2} \cong 4kT \frac{4}{5} \frac{Q}{\omega_0 C}$$

$$\Delta C_{xmin} \text{Sensitivity} \cdot i_{in} = \sqrt{\text{Noise} \cdot B_m} \quad \Delta C_{xmin} \cong \frac{\sqrt{\text{Noise} \cdot B_m}}{\frac{4\sqrt{5}}{25} \frac{Q}{C} \cdot R i_{in}}$$

$$\Delta C_{xmin} = \frac{25}{4\sqrt{5}} \sqrt{\text{Noise} \cdot B_m} \frac{C}{Q V_{ac}}$$

Example:

$$Q = 100, f_0 = 220\text{MHz}, B_m = 1\text{s}, C = 5\text{pF}, V_{ac} = 0.1\text{V}$$



$$\Delta C_{x\min}^{\text{ris}} = 20\text{zF}$$

Comparison with V/I:

$$\Delta C_{x\min}^{\text{ris}} \approx \frac{5}{2} \sqrt{4kT \frac{1}{\omega_0 C Q} B_m \frac{C}{V_{ac}}}$$

$$\Delta C_{x\min}^{V/I} \approx \frac{\sqrt{\overline{e_n^2} \omega^2 C_T^2 B_m}}{\omega V_{ac}} \cong \frac{\sqrt{4kT \frac{1}{g_m} C_T^2 B_m}}{V_{ac}}$$

$$\Delta C_{x\min}^{\text{ris}} = \frac{5}{2} \Delta C_{x\min}^{V/I} \sqrt{\frac{g_m}{\omega_0 C Q}}$$

$$g_m = 10\text{mS} \rightarrow Q > 10$$



Resonance Pros and Cons

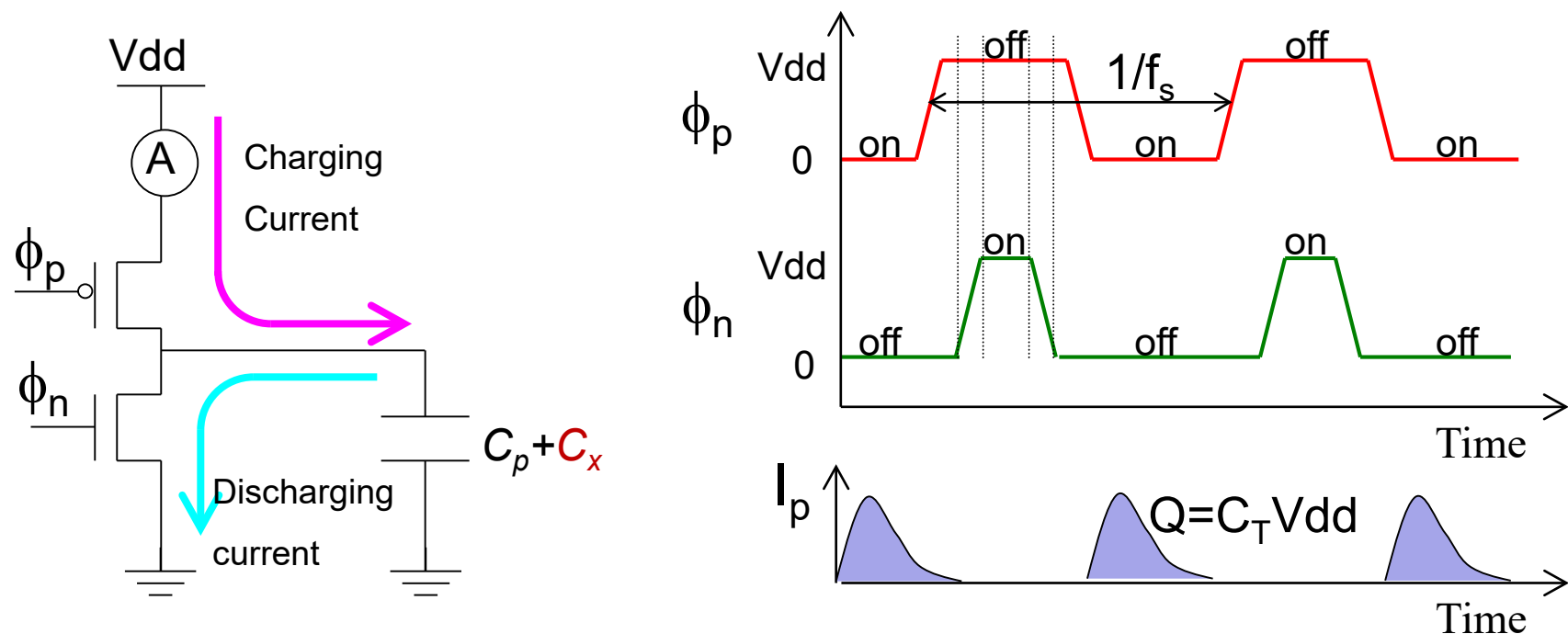
Pro:

- Ultra high sensitivity

Cons:

- Suitable only for capacitive sample (high Q)
- Limited frequency range
- Difficult to measure the absolute impedance
- High frequency operation more complicated instrumentation

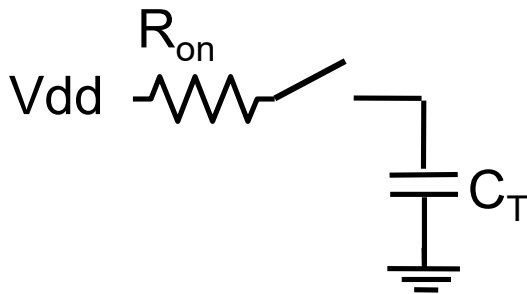
Charge-Based Measurement



- Different charge and discharge paths
- Periodically charge at a known voltage V_{dd} (full discharge)
- $I_{\text{average}} = (C_p + C_x) f_s V_{dd} = C_T f_s V_{dd}$
- Easy to implement and measure

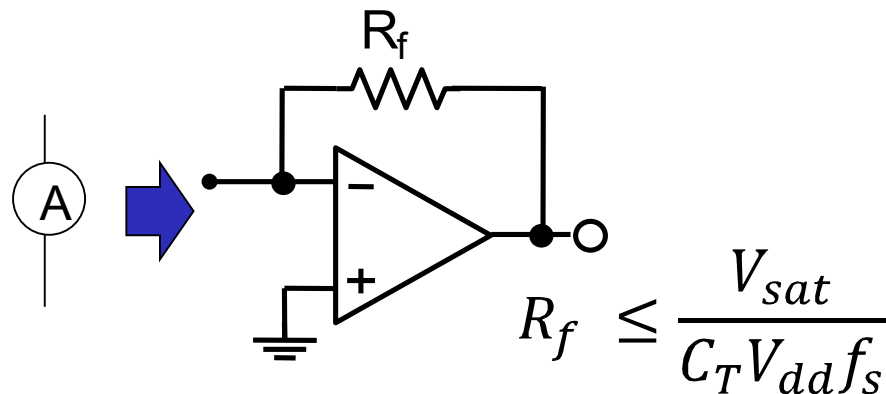


Noise kTC:



$$\overline{Q_c^2} = kTC_T$$

$$Noise = \sqrt{\overline{i_{eq}^2} B_m + 2kTC_T f_s B_m}$$



$$Noise \cong \sqrt{4kT \frac{C_T V_{dd} f_s}{V_{sat}} B_m + 2kTC_T f_s B_m}$$

$$Noise \cong \sqrt{2kTC_T f_s B_m}$$



$$\Delta C_{xmin}^{CBCM} = \frac{Noise}{f_s V_{dd}} \cong \frac{\sqrt{2kTC_T B_m}}{V_{dd} \sqrt{f_s}}$$

Example:

$$V_{DD} = 100\text{mV}, B_m = 1\text{s}, C_T = 5\text{pF}, f_s = 10\text{MHz} \rightarrow \Delta C_{xmin} = 0.6\text{aF}$$

Comparison with current sensing:

$$\Delta C_{xmin}^{V/I} \approx \frac{\sqrt{\overline{e_n^2} \omega^2 C_T^2 B_m}}{\omega V_{ac}} \cong \frac{\sqrt{4kT \frac{1}{g_m} C_T^2 B_m}}{V_{ac}}$$

Same $V_{dd} = V_{ac}$ and B_m : $\Delta C_{xmin}^{CBCM} = \Delta C_{xmin}^{V/I} \sqrt{\frac{g_m}{2C_T f_s}}$

$g_m = 10\text{mS}$

 $f_s = 1\text{GHz}!!!$



Pros:

- Current reader operates at low frequency
- Compatible with CMOS integration

Cons:

- Mostly for capacitance measurements
- If R is present $i = 0.5V_{dd}/R + C_T f_s V_{dd}$ change f_s
- Charge injection, clock feedthrough, $R_{DS,on}$ limit the accuracy
- Spectroscopy is impossible



- **Current sensing:**
 - Excellent control of experimental conditions
 - High sensitivity
 - Wide spectrum
- **Lock-in** to maximize SNR
- **FFT** for speed (powerful DSP)
- **Resonant techniques** high sensitivity but not versatile and more complicated to implement
- **CBCM** simple (IC) but only for capacitance, no spectroscopy